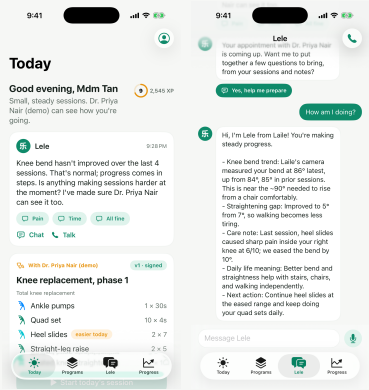
Laile

Home physio that counts every rep and checks in.

Tencent Cloud AI CAN DO IT Hackathon Singapore 2026
Healthcare track • Challenge Statement 2

Charles Andrew Lim



Physio happens at home, where nobody sees it

50%

of people in rehab may not do their home exercise [1]

21.4%

of Singapore citizens are 65 or older (2026), a super-aged society [3]

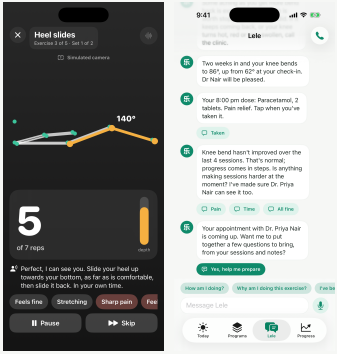
Why people stop (systematic review of 20 studies [2]): low confidence, low mood, worry, pain during exercise, no support, too many barriers.

What a physio sees between visits: nothing. A checkbox can't tell a done exercise from a skipped one, or one stopped because it hurt.

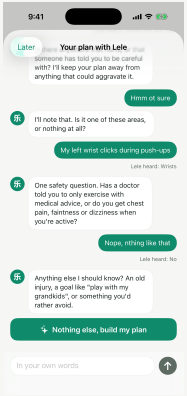
What the brief asks for: support that is empowering, personalised, longitudinal, safe, strictly bounded and honest about uncertainty.

Laile in one breath

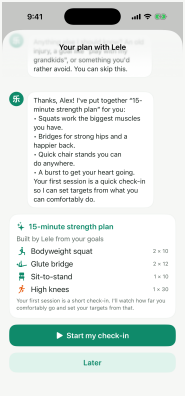
- **Programs are the unit.** A physio's program (joined with their code), a plan Lele builds from your goals, or a calisthenics program.
- **The camera counts.** Reps, holds, knee bend and arm lift, measured on the phone. Video stays there.
- **Lele checks in.** A coach that talks you through sessions, asks how you are each morning, reminds, notices, remembers what hurt.
- **The physio stays on top.** Live sessions, a daily list, approvals, signatures, discharge.
- **Code decides, the model talks.** Safety and every number come from tested rules.
- **iPhone or any browser.** Same account, same sessions, same Lele.



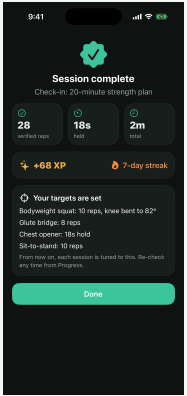
Start: tell Lele your goals, or bring a physio's code



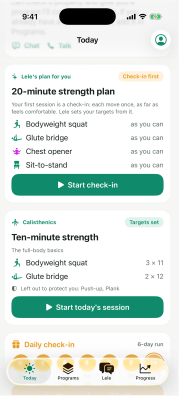
Goals in your own words; options are suggestions



Lele's plan leaves out what your wrists can't take

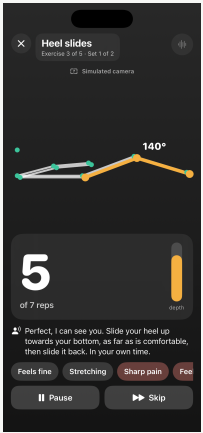


First session is a check-in that sets targets



Limits follow you into every program

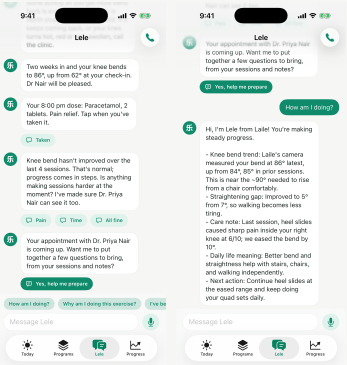
Sessions: the camera counts, Lele coaches



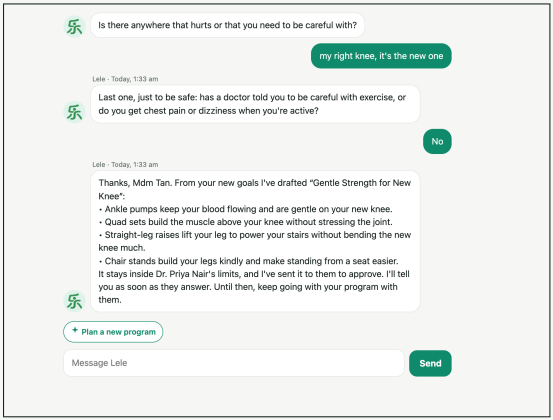
- Apple Vision on iPhone, MediaPipe in the browser: joints are tracked on the device; a rep counts only at the target angle, a hold only while in position.
- Lele takes turns like a conversation: explains, says *"In your own time"*, counts, asks *"How did that feel?"* between sets.
- Say "it's pulling" or "sharp pain" any time. Fixed rules decide: carry on, stop the set, stop, or get help.
- Something sore? Lele notes it and makes that exercise easier next time, then eases back after two comfortable sessions.
- Can't see you? No count: *"I can't see you clearly, so I'm not counting that."*

Lele takes initiative, and you can talk back

- **Morning check-in** in physio. “Quite swollen this morning” makes today’s session gentler and tells the physio.
- **Medicine reminders** at the doctor’s times, as notifications too; “What’s this for?” on every dose.
- **Missed days:** one nudge, then “what’s getting in the way?”
- **Plateaus and visit prep** from real measurements.
- **Chat or call** Lele, built on Tencent Cloud ADP with a curated knowledge base.
- **Never nags:** quiet 9 pm to 7 am, three conversations a day at most.

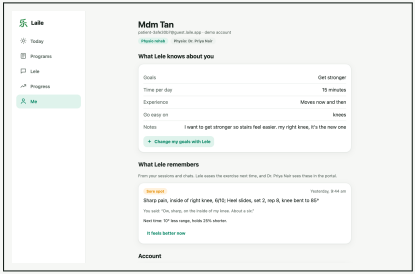


Talk to Lele to change your goals

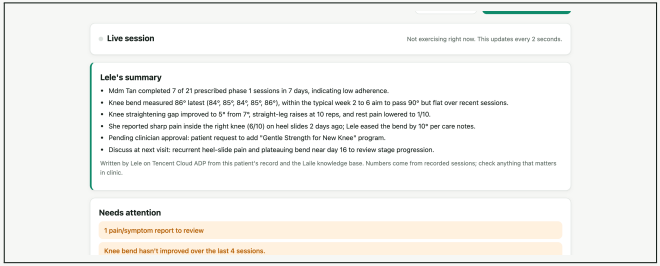


- Tap *Plan a new program*, or just say "my goals have changed".
- Five questions: goals, minutes, experience, what hurts, a safety check. Tap an answer or use your own words.
- Lele on **Tencent Cloud ADP** drafts the program from the knowledge base, using only Laile's vetted exercises. Code checks it against the person's limits.
- **No physio:** it replaces Lele's old plan and starts with a check-in.
- **Physio patient:** it goes to the physio to approve. Nothing starts before a yes.

Lele learns each person, and briefs the physio



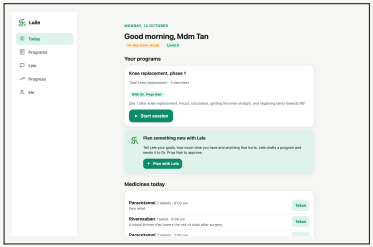
What Lele knows and remembers, on the person's Me screen



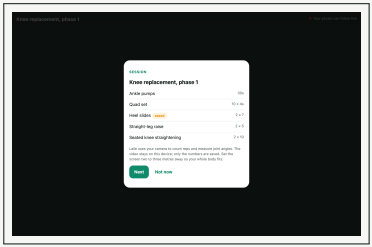
Lele's summary on the physio's patient page

- Every chat and session adds to one record: goals, time, sore spots in their words, what was eased, measurements, requests.
- Lele reads it before every reply. For the physio, ADP writes a 4 to 6 line brief from that record plus the knowledge base ("typical week 2 to 6 aim to pass 90°"). The numbers come from code.

Laile in any browser, no install

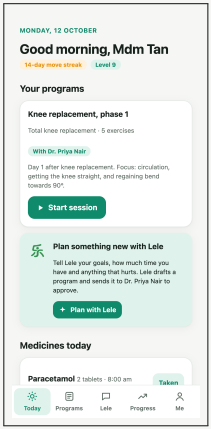


Today on a laptop



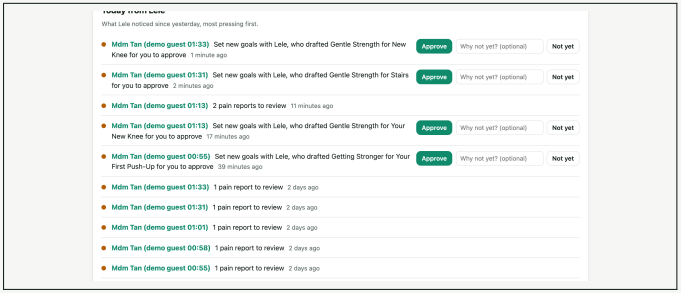
A webcam session; heel slides eased by Lele

- Same account, programs, Lele and progress as the iPhone app.
- MediaPipe pose in the browser and a port of Laile's session engine, tested against the same plan: web sessions count and report like phone sessions.
- Lele's ElevenLabs voice; the physio can watch live.



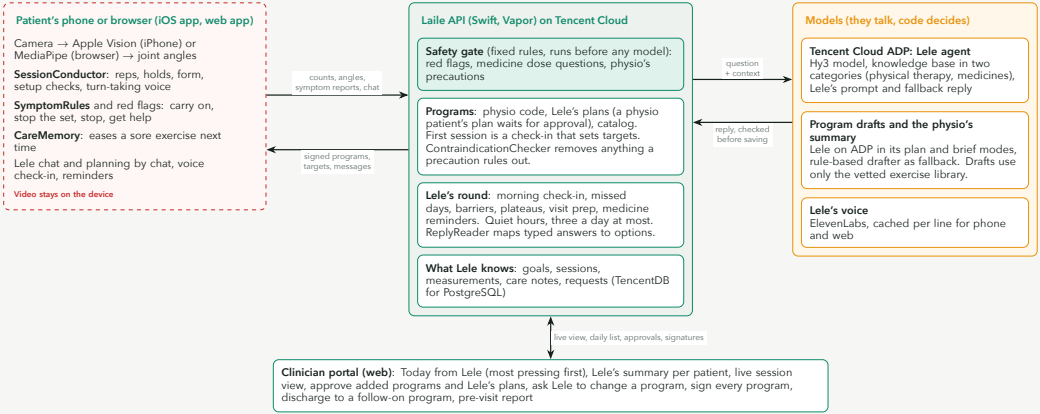
43-156-108-79.sslip.io/app

The physio stays on top



- **Today from Lele**, most pressing first: red flags, worse this morning, pain reports, requests, missed days.
- **Live view** of a session: exercise, reps, measured angle.
- **Approve** extra programs and **Lele's plans**; **ask Lele** to change one in plain words; **sign** every version.
- **Discharge** hands the patient to a follow-on program, limits kept.

Architecture and trust boundary



How the AI is used, and kept safe

Job	Done by	Kept safe by
Conversation and explanations	Lele on **Tencent Cloud ADP** (Hy3, 12-document knowledge base)	Real context sent with each message; red flags and dose questions answered by rules first
Remembering what hurt	Care notes	Easing computed by code; physio sees every note
Program drafts	Lele on **ADP** (plan mode, knowledge base); rule-based fallback	Vetted library only, precautions enforced, physio signs or approves
Brief for the physio	Lele on **ADP** (brief mode) with the record and knowledge base	Numbers and flags computed by code and passed in
Understanding replies	Keyword rules	"Lele heard: …"; asks again if unclear
Seeing movement	Apple Vision (iPhone), MediaPipe (browser)	Video stays on the device

- Dose questions: fixed referral to the doctor or pharmacist. Unknown medicine: *"I'm not able to identify that medication, please check with your pharmacist."*
- 158 automated tests cover counting, safety rules, care notes, Lele's round, planning by chat, the server flows and the web session engine.

Built with

Tencent Cloud and Tencent tools

ADP: Lele agent on the Hy3 model with a 12-document knowledge base: chat, program drafts, the physio's summary

Lighthouse: hosts the API, web app and clinician portal (Docker, Caddy, HTTPS)

Hunyuan (TokenHub): optional direct model

CodeBuddy / WorkBuddy: safety and ADP audits, safety fixes (Hy4)

Miora: logo, cover, app icon and the Lele mascot (Hy Image 3.5)

iPhone and web apps

Swift 6, SwiftUI: feature modules in LaileKit

Apple Vision, Speech: body pose for reps, holds and angles; spoken symptoms

Web app: plain JavaScript, MediaPipe pose in the browser, the same API

ElevenLabs: Lele's voice on phone and web, cached per line

TestFlight: public link for testers

Server and shared code

Vapor (Swift): API, Leaf clinician portal, live session view over WebSocket

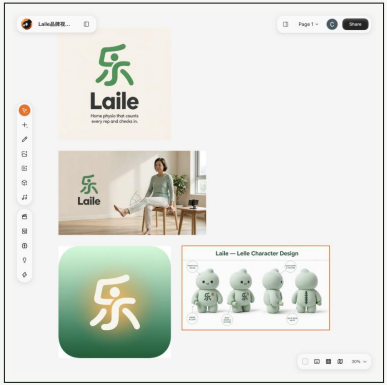
Fluent: SQLite, or TencentDB for PostgreSQL

LaileCore: one Swift package for safety rules, programs, care notes and Lele's round, shared by phone and server

158 automated tests: 121 core, 32 server, 5 web

Tooling: XcodeGen, Docker Compose, Tectonic

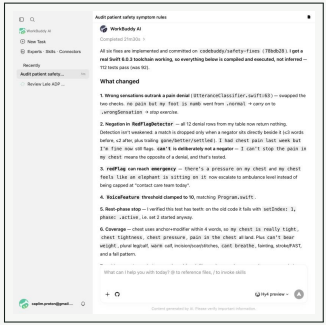
Made with Tencent's own tools



Miora: logo, cover, app icon, Lele mascot



WorkBuddy: safety audit with file and line findings



WorkBuddy: six safety fixes, 112 tests passing

- **CodeBuddy** found that "no pain but my foot is numb" was answered with "carry on" and that "I haven't taken it yet" was logged as a dose; its fixes were reviewed and merged.
- **ADP** runs Lele (chat, drafts, physio summaries); **Lighthouse** hosts the server, web app and portal.

Fit with Challenge 2

Medication understanding	“What’s this for?” on every dose, from the medicines knowledge base
Medication management	Reminders at the doctor’s times; one tap or a typed reply marks a dose taken
Adherence support	Camera-verified sessions, check-in targets, forgiving streaks, barrier questions, new plans by chat
Health tracking	Knee bend, straightening gap, arm lift, reps, holds and pain against baseline
Preventive health	Precautions follow you; plateau and missed-day flags; follow-on programs
Healthcare preparation	Visit prep questions; pre-visit report and Lele’s summary for the physio
Safe, bounded, honest	Rule-based red flags and referrals; no count without a clear view

Why it can last

Clinics
Verified adherence, live sessions and a prioritised daily list without extra visits. Proposed: a monthly fee per active patient.

People
Discharge leads to a calisthenics or mobility program with limits carried over. Consumer subscription for people without a physio.

A loop
Everyday users with aches are pointed to a physio; physio patients become everyday users afterwards.

Cheap to run
Pose estimation on the device; model calls only for conversation, drafts and briefs; voice lines cached and paid for once.

Where it stands

Working now

- iOS app on TestFlight with a public link
- Web app with webcam sessions, no install
- Server, portal and web app on Tencent Cloud Lighthouse
- Lele on Tencent Cloud ADP: chat, plans, physio summaries
- Camera pipeline tested with simulated skeletons
- 158 automated tests

Next

- TencentDB for PostgreSQL
- Testing with people at home: camera placement, lighting, voice across a room
- Knowledge base review by a physiotherapist and a pharmacist
- Pilot with one clinic

Source: github.com/LimCharles/laile • Try Lele: adp.tencentcloud.com/webim_exp/#/chat/nZbZZz
Portal: 43-156-108-79.sslip.io/portal (clinician@laile.demo / laile-demo-2026)
Web app: 43-156-108-79.sslip.io/app • iPhone app: testflight.apple.com/join/WuymKzUH

References

[1] Argent R, Daly A, Caulfield B. Patient involvement with home-based exercise programs: can connected health interventions influence adherence? *JMIR mHealth uHealth* 2018;6(3):e47.

[2] Jack K, McLean SM, Moffett JK, Gardiner E. Barriers to treatment adherence in physiotherapy outpatient clinics: a systematic review. *Manual Therapy* 2010;15(3):220-228.

[3] National Population and Talent Division. *Population in Brief 2026*. Singapore, September 2026.